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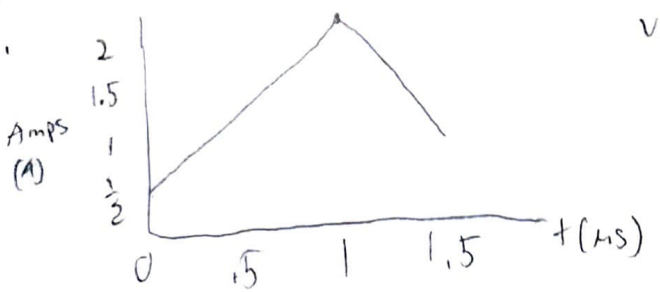
EEN 164 HW2

1.

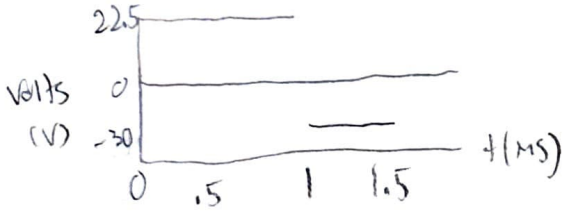


$v = L\frac{di}{dt}$

$\rightarrow 15\,mH\left(\frac{2-\frac{1}{2}}{(1-0)\times10^{-3}}\right) = 22.5V$

$15\,mH\left(\frac{1-2}{(1.5-1)\times10^{-3}}\right) = -30V$

a.



b. Not steady state. The avg voltage over the period does not equal 0 (value of A at the end of period != what it was at the start). $22.5(1) + (-30)(.5) = 7.5$ not 0

2.

(i) $P_{in} = Power_{disp} + P_{out}$ $\quad P_{out} = 2W$

$\frac{2}{.9} = 2.\bar{2}$

$2.\bar{2} = Power_{disp} + 2$

$\boxed{Power_{disp} = 222mW}$

(ii) $P_{in} = Power_{disp} + .88\,P_{in}$

$.12\,P_{in} = Power_{disp} = .222$

$P_{in} = 1.852W$

$\boxed{P_{out} = .88(P_{in}) = 1.630\,W}$

3. $v(t) = 120\sqrt{2}\cos(2\pi 60t)$

	i(t)	Elt	Power	PF	THD
1	$20\sqrt{2}\cos(2\pi 60t) = 20\sqrt{2}\angle 0$	resistor	2400 W	1	0
2	$10\sqrt{2}\cos(2\pi 60t - 40°) = 10\sqrt{2}\angle -40$	resistor + inductor	919.253 W	.766	0
3	$12\sqrt{2}\cos(2\pi 60t + 20°) = 12\sqrt{2}\angle 20$	resistor + capacitor	1353.157 W	.940	0
4	$1 + 6\cos(2\pi 60t - 20) + 3\cos(4\pi 60t - 15) + \cos(6\pi 60t - 25)$	some nonlinear component	478.413 W	.814	$\frac{\sqrt{3}}{3}$

$\frac{120\sqrt{2}\cdot 20\sqrt{2}}{2} = 2400$ $\quad \frac{120\sqrt{2}\cdot 10\sqrt{2}}{2}\cos(0-(-40))$ $\quad \frac{120\sqrt{2}\,12\sqrt{2}}{2}\cos(0-20)$ $\quad \frac{120\sqrt{2}\cdot 6}{2}\cos(0+20) =$

1200 $\quad = 919.253$ $\quad 1440 * .940 = 1353.157$ $\quad 478.413$

$I_{RMS} = \sqrt{1^2 + \frac{36}{2} + \frac{9}{2} + \frac{1}{2}}$

$= 4.699$

$$\frac{478.413}{\frac{120\sqrt{2}\cdot 4.899}{2}} = .814 \qquad \frac{6}{\sqrt{2}} = 4.243$$

$$\sqrt{\frac{4.899^2 - 4.243^2}{4.243^2}} = \frac{\sqrt{3}}{3}$$

4.

$$v(t) = 2 + 5\cos(2\pi 60t) - 3\cos(4\pi 60t + 45)$$

$$i(t) = 1.5 + 2\cos(2\pi 60t + 20) + 1.1\cos(4\pi 60t - 20)$$

a.

$$V_{rms} = \sqrt{2^2 + \frac{25}{2} + \frac{9}{2}} = 4.583V$$

$$I_{RMS} = \sqrt{1.5^2 + \frac{4}{2} + \frac{1.21}{2}} = 2.203A$$

$$P = 2(1.5) + \frac{5(2)}{2}\cos(0-20) + \frac{(-3)1.1}{2}\cos(45-(-20))$$

$$3 + 4.698 + -.697 = \boxed{7.001W}$$

Not asked

PF: $\frac{7.001}{4.583 * 2.203} = .693$